Math 141, F11 Name: Answer

Quiz 4 Section: SSN: xxx-xx-

Instructions: There are totally 3 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1.(2 points) Compute

$$\lim_{x \to -\infty} x^2 + x^3$$

$$= \lim_{x \to -\infty} x^2(1+x) = -\infty$$

Problem 2.(4 points) Calculate $f'(2) = \lim_{h \to 0} \frac{f(2+h)-f(2)}{h}$ for $f(x) = \dfrac{1}{x+1}$.

Solution: $f'(2) = \lim_{h \to 0} \dfrac{f(2+h)-f(2)}{h}$

$$= \lim_{h \to 0} \frac{\frac{1}{(2+h)+1} - \frac{1}{2+1}}{h}$$

$$= \lim_{h \to 0} \frac{\frac{1}{3+h} - \frac{1}{3}}{h} = \lim_{h \to 0} \frac{\frac{3-(3+h)}{3(3+h)}}{h}$$

$$= \lim_{h \to 0} \frac{\frac{-h}{3(3+h)}}{h}$$

$$= \lim_{h \to 0} -\frac{1}{3(3+h)} = -\frac{1}{9}$$

Problem 3.(4 points) Let $f(x) = 2x^2 - 1$, find its derivative $f'(x)$ by evaluating $\lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$.

Solution: $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$

$$= \lim_{h \to 0} \frac{[2(x+h)^2 - 1] - [2x^2 - 1]}{h}$$

$$= \lim_{h \to 0} \frac{[2x^2 + 4xh + 2h^2 - 1] - [2x^2 - 1]}{h}$$

$$= \lim_{h \to 0} \frac{4xh + 2h^2}{h}$$

$$= \lim_{h \to 0} (4x + 2h)$$

$$= 4x$$